

TYCHE R2-T1

Final Controlled Values Register

Single source of truth for final design report reconciliation and submission QA

Control item	Released state
Aircraft architecture	Tyche R2-T1 - twin segregated parallel-hybrid STOL aircraft
Geometry authority	G1.3 analysis-ready geometry; 10.750 m fuselage
Structural / mass baseline	S1.0 report mass and structural state, updated by the executed CG / control closure
Aerodynamic production model	G5 thin lifting-system branch: MainWing + HTail + VTail + lift struts
Final engineering state	FG8.0-FG8.4 closed; VMC7.4 conditional analytical closure; final uncertainty register complete
Register release	CVR-1.0 - 27 August 2026
Primary purpose	Every numerical claim in the final report must agree with this register or be explicitly labelled as a sensitivity / historical value.

Authority and precedence rules

- Latest controlled analysis supersedes earlier screening studies, preliminary architecture work, stale report text, and chronological intermediate conclusions in the Analysis Log.
- A value tagged FINAL is the preferred report value. REPORT BASIS means the value remains a deliberate design/performance assumption rather than a directly validated quantity.
- CONDITIONAL and HOLD items must retain their stated qualification in the final report; they must not be written as unconditional compliance.
- SUPERSEDED values may appear only when explicitly described as historical or diagnostic evidence.
- No further OpenVSP/VSPAERO, mesh, wake, fairing, powered-flow or optional simulator work is authorised before submission unless report integration exposes a direct arithmetic or configuration-control error.

Critical report-integration warnings

- Do not use Geometry G1.1 or the 9.70 m fuselage as the released geometry. Use G1.3 and 10.750 m.
- Do not describe $CD_0 = 0.0283$ as OpenVSP-validated. It is a deliberately conservative performance basis above the executed PDT build-up.
- Do not use FS 3.60-3.90 m as the operational CG envelope. Use the actual loading cloud and U20 screening boundary in this register.
- Do not use the earlier 38.73/39.09 kn VMC headline values. Use the atmosphere-consistent 38.97/39.37 kn EAS results and retain the 40.97 kn 90%-travel robustness failure.
- FG8 Fowler results are longitudinal-control surrogates; they do not validate the final multi-element high-lift CL_{max} or landing drag polar.

1. Controlled RFP and Mission Basis

ID	Quantity / rule	Controlled value	Use / status	Source
RFP-01	Maximum take-off mass	2,000 kg	HARD LIMIT	S04/S06
RFP-02	Occupants	1 pilot + 5 passengers = 6 persons	HARD REQUIREMENT	S04/S06
RFP-03	Standard person mass	86 kg/person; 516 kg total occupants	CONTROLLED ASSUMPTION	S04/S06
RFP-04	Additional payload	200 kg, simultaneous with passenger mission	HARD REQUIREMENT	S04/S06
RFP-05	Total people + additional payload	716 kg	FINAL arithmetic basis	S04/S06
RFP-06	Design stage	505 km	FINAL; >500 km requirement	S04/S06
RFP-07	Final reserve	30 min VFR, fuel based	FINAL mission basis; no routine battery cruise credit	S04/S06
RFP-08	Cruise point	130 kn at 2,600 m	FINAL; lower bound of 130-150 kn band	S04/S06
RFP-09	Take-off distance limit	<=200 m to 15 m obstacle	HARD LIMIT	S04/S06
RFP-10	Landing distance limit	<=150 m from 15 m obstacle	HARD LIMIT	S04/S06
RFP-11	Hot-high field condition	610 m (2,000 ft), ISA+20 C; no beneficial wind	FINAL operating screen	S04/S06
RFP-12	Stall-speed requirement	<50 kn	HARD LIMIT	S04/S06
RFP-13	Wing-loading limit	<=90 kg/m ² at MTOM	HARD LIMIT	S04/S06
RFP-14	Service ceiling	>=3,000 m	HARD REQUIREMENT	S04/S06
RFP-15	Structural limit / ultimate basis	+3.8 / -1.5 g limit; 1.5 ultimate factor	FINAL structural basis	S04/S07

2. Final Configuration and Geometry

ID	Quantity	Controlled value	Use / status	Source
GEO-01	Geometry release	G1.3 analysis-ready	FINAL - supersedes G1.1	S05
GEO-02	Fuselage overall length	10.750 m	FINAL	S03/S05
GEO-03	Fuselage maximum width / height	~1.55 m / ~1.70 m	REPORT GEOMETRY BASIS	S04/S05
GEO-04	Wing reference area, S	30.000 m ²	FINAL	S03/S05
GEO-05	Wing span, b	16.400 m	FINAL	S03/S05
GEO-06	Aspect ratio	8.96533	FINAL derived	S03/S05

ID	Quantity	Controlled value	Use / status	Source
GEO-07	Mean aerodynamic chord	1.881 m (G1.3 exact ~1.880663 m)	FINAL report value	S03/S05
GEO-08	Taper ratio	0.55	FINAL geometry	S05
GEO-09	Root / tip chord	2.360346 m / 1.298190 m	FINAL derived geometry	S05
GEO-10	Wing section / thickness	NASA LS(1)-0417 / 17% c	FINAL design basis	S04/S05
GEO-11	Wing rigging	+2 deg root incidence; 2 deg washout	FINAL design basis	S04/S05
GEO-12	MAC leading edge	FS 3.050 m	FINAL datum	S03/S05
GEO-13	Wing quarter-chord station	FS 3.520166 m	FINAL derived datum	S03/S05
GEO-14	Wing-root LE / TE stations	FS 2.930079 / 5.290425 m	FINAL geometry	S03/S05
GEO-15	HT aerodynamic-centre station	FS 9.420166 m	FINAL geometry	S03/S05
GEO-16	Wing-to-tail c/4 arm	5.90 m	FINAL retained design arm	S03/S05
GEO-17	Propellers	2 x 2.00 m diameter at y = +/-1.60 m	FINAL architecture	S02/S05
GEO-18	Landing-gear stations / track	Nose FS 0.750 m; Main FS 4.290 m; wheelbase 3.540 m; track 3.900 m	FINAL geometry	S02/S05
GEO-19	High-lift design geometry	Slat 0.70-5.95 m; Fowler 0.85-5.85 m semispan; aileron 5.95-8.00 m	REPORT BASIS; not direct CLmax validation	S04/S07
GEO-20	High-lift deflections	Slat 15 deg TO / 25 deg LDG; Fowler 12 deg TO / 35 deg LDG	REPORT BASIS / control-surrogate geometry	S04/S02

3. Propulsion and Energy System

ID	Quantity	Controlled value	Use / status	Source
PWR-01	Architecture	Two segregated parallel-hybrid channels	FINAL	S04
PWR-02	Combustion engines	2 x Rotax 916 iSc3 B; 117 kW take-off and 101 kW continuous each	FINAL COTS basis	S04/S07
PWR-03	Engine mass basis	~86 kg dry each; 128 kg installed each in report mass ledger	FINAL mass statement basis	S04/S07
PWR-04	Electric machines	2 x Textron/Pipistrel E-811-268MVL; 57.6 kW max (90 s), 49.2 kW continuous each	FINAL COTS basis	S04/S07
PWR-05	Electric-machine installed allowance	36 kg per channel	FINAL mass statement basis	S04/S07
PWR-06	Propeller type	2.0 m constant-speed, full-feathering, reversible-pitch, fixed axis	FINAL architecture; exact supplier map OPEN	S04/S07

ID	Quantity	Controlled value	Use / status	Source
PWR-07	Take-off shaft power, total	320.25 kW $\approx$ 320.2 kW	FINAL report calculation	S04/S07
PWR-08	VMC live shaft-power bound	160.13 kW per live channel at 2279 RPM	FINAL analytical VMC bound	S02
PWR-09	Continuous cruise availability	196 kW total	FINAL report performance basis	S04
PWR-10	ESS	11.0 kWh nameplate; 8.8 kWh usable; 71.1 kg installed	FINAL mass/energy basis	S04/S07
PWR-11	Battery dispatch philosophy	No routine cruise draw; take-off/initial-climb/go-around augmentation and 5 min OEI support	FINAL operating philosophy	S04
PWR-12	Propeller rotation architecture	Counter-rotation retained as favourable architecture sensitivity; exact supplier implementation not frozen	OPEN hardware confirmation	S01/S02

4. Mass, Payload and CG Envelope

ID	Quantity / loading state	Controlled value	Use / status	Source
MAS-01	Operating empty mass	1,327.0 kg	FINAL mass statement	S04/S03
MAS-02	Mission fuel allocation	109.8 kg	FINAL 505 km mission basis	S04
MAS-03	Full RFP mission mass	2,152.8 kg	FAIL: exceeds MTOM by 152.8 kg	S04/S06
MAS-04	OEW required for full RFP mission	1,174.2 kg	CLOSURE THRESHOLD; 152.8 kg below current OEW	S04/S06
MAS-05	Max people + payload at 505 km / MTOM	563.2 kg	FINAL delivered-capability point	S04
MAS-06	Cargo with six 86 kg occupants at 505 km	47.2 kg	FINAL delivered-capability point, not RFP compliance	S04
MAS-07	Full 716 kg payload with zero fuel	2,043 kg	FAIL: still above MTOM	S04
CG-01	Geometry-based OEW CG	FS 3.42993 m = 20.20% MAC	FINAL conceptual mass-arm result	S03
CG-02	Nominal loading-cloud extent	FS 3.34618 to 3.70312 m	FINAL nominal cloud	S03
CG-03	Most-aft nominal static margin, U20	17.16% MAC at FS 3.70312 m	PASS	S03
CG-04	Independent-uniform 95% aft screen	FS 3.76080 m; 14.09% MAC	PASS	S03
CG-05	All-at-bounds aft screen	FS 3.91324 m; 5.99% MAC	HOLD / FAIL operational release	S03
CG-06	Fuselage-corrected neutral point	FS 4.02894 m = 52.04% MAC	FINAL central analytical NP	S03

ID	Quantity / loading state	Controlled value	Use / status	Source
CG-07	Conservative U20 neutral point / 10%-MAC aft screen	FS 4.02588 m / FS 3.83778 m	FINAL report screening boundary	S03
CG-08	Forward take-off control state	1,608.8 kg; FS 3.37374 m	FINAL FG8.1 case	S02/S03
CG-09	Forward landing/go-around control state	1,519.8 kg; FS 3.35170 m	FINAL FG8.2/FG8.3 case	S02/S03
CG-10	Operational loading restriction	No loading beyond FS 3.83778 m pending CAD-quality mass-arm confirmation	CONDITIONAL / HOLD boundary	S02/S03

5. Aerodynamics and High-Lift

ID	Quantity	Controlled value	Use / status	Source
AER-01	Production aerodynamic model	Thin lifting-system: MainWing + HTail + VTail + lift struts; propulsion off	FINAL production source	S01/S08
AER-02	Mixed thin/thick body model	Rejected after geometry repair because complete assembly remained physically invalid	DO NOT use coefficients	S01/S08
AER-03	CA-G3 clean CL0	0.43349	FINAL clean-polar value	S01
AER-04	CA-G3 clean CLalpha	5.6161 rad ⁻¹	FINAL broad clean-polar fit	S01
AER-05	CA-G3 zero-lift alpha	-4.422 deg	FINAL clean-polar value	S01
AER-06	CA-G3 surface induced-drag factor, k	0.044608	FINAL working surface fit	S01
AER-07	CA-G3 fitted Oswald efficiency, e	0.7959	FINAL working surface fit	S01
AER-08	Nominal clean total polar	CD = 0.0216437 + 0.0446077 CL ²	VALIDATION/interpretation polar	S01
AER-09	Combined-adverse clean polar	CD = 0.0232097 + 0.0469719 CL ²	ROBUSTNESS interpretation	S01
AER-10	Nominal / combined-adverse L/Dmax	16.09 / 15.14	VALIDATION performance inputs	S01
AER-11	Wing-only analytical CLalpha	4.837 rad ⁻¹	ANALYTICAL reference	S02/S07
AER-12	MainWing-only VSPAERO CLalpha	4.88866 rad ⁻¹	VALIDATED comparison; +1.07% vs analytical	S03/S02
AER-13	Complete lifting-system local CLalpha	5.61880 rad ⁻¹	FINAL stability derivative; not wing-only	S01/S03
AER-14	PDT geometry CD0	0.018985	EXECUTED lower build-up	S02
AER-15	PDT operational CD0	0.021501	EXECUTED operational build-up	S02
AER-16	PDT adverse upper CD0	0.023067	EXECUTED upper build-up	S02

ID	Quantity	Controlled value	Use / status	Source
AER-17	Performance CD0 basis	0.0283	REPORT BASIS - deliberately conservative; do not call validated	S02/S04
AER-18	Unpowered take-off CLmax design basis	2.40	REPORT BASIS; analytical high-lift build-up	S04
AER-19	Powered take-off CLmax design basis	2.76 (= 2.40 + bounded 0.36 slipstream credit)	REPORT BASIS; powered-lift correlation OPEN	S04
AER-20	Unpowered landing CLmax design basis	3.30	REPORT BASIS; high-lift correlation OPEN	S04
AER-21	Induced-drag uncertainty	~5.25% k and ~5.54% e Fine-to-XF sensitivity; ~1.26% wake-node sensitivity	CARRY as numerical uncertainty	S01/S02
AER-22	Surface / Trefftz drag discrepancy	Material, ~34% at alpha=4 deg in clean campaign	DOCUMENT; use Trefftz CDi,w for convergence	S01

6. Stability, Control and Minimum Control Speed

ID	Quantity	Controlled value	Use / status	Source
STB-01	CYbeta	-0.42565 rad ⁻¹	FINAL G5 multi-point derivative	S01
STB-02	Clbeta	-0.071802 rad ⁻¹	FINAL; manual multi-point value preferred	S01
STB-03	Cnbeta	+0.13247 rad ⁻¹	FINAL G5 derivative	S01
STB-04	Steady rate damping	Clp = -0.55079; Cm _q = -19.25475; Cnr = -0.11230	FINAL stability evidence	S01
CTL-01	Elevator effectiveness	Cm _{delta_e} = -2.2607 rad ⁻¹ ; CL _{delta_e} = +0.7832 rad ⁻¹	FINAL; positive report convention = TE-down	S01
CTL-02	Aileron effectiveness	Cl _{delta_a} = -0.081393; Cn _{delta_a} = +0.004979; CY _{delta_a} = +0.006868 rad ⁻¹	FINAL	S01
CTL-03	Rudder effectiveness	Cn _{delta_r} = +0.11056; CY _{delta_r} = -0.29803; Cl _{delta_r} = -0.02735 rad ⁻¹	FINAL	S01
STB-05	Lifting-system Cm _{alpha}	-1.17352 rad ⁻¹	FINAL G5 local derivative	S01/S03
STB-06	Munk-Multhopp fuselage Cm _{alpha} correction	+0.34028 rad ⁻¹ , destabilising	FINAL analytical correction	S03
STB-07	Corrected complete analytical Cm _{alpha}	-0.83324 rad ⁻¹	FINAL fuselage-corrected value	S03
STB-08	Corrected neutral point	FS 4.02894 m = 52.04% MAC	FINAL central value	S03

ID	Quantity	Controlled value	Use / status	Source
CTL-04	Forward-CG hot-high rotation authority	~15 deg physical TE-up elevator required	PASS / CLOSED	S02
CTL-05	Minimum nose-up design travel with 10% geometric reserve	>=17 deg physical TE-up	FINAL design requirement	S02
CTL-06	Forward landing steady trim	~4.08 deg physical TE-up; trim alpha ~4.54 deg	PASS / CLOSED	S02
CTL-07	Demonstrated flare reserve	~5.92 deg additional tested TE-up movement; ~2.85 kN m additional nose-up moment at tested -10 deg GUI state	PASS conceptual screen	S02
CTL-08	Go-around flap-transition screen	35 -> 20 -> 12 deg Fowler: smooth lift/Cm trend, no pitch-control reversal	PASS / CLOSED; quasi-steady only	S02
VMC-01	VMC release atmosphere	Sea-level standard rho = 1.225 kg/m ³ ; EAS reference rho0 = 1.225 kg/m ³	FINAL bookkeeping basis	S02
VMC-02	VMC hot-high sensitivity atmosphere	610 m ISA+20 C; rho = 1.079 kg/m ³	SENSITIVITY	S02
VMC-03	Physical control-stop basis	Aileron +/-22 deg; rudder +/-35 deg; beta <=5 deg; phi <=5 deg	CONDITIONAL hardware basis	S02
VMC-04	Near-stall effectiveness debit	75% aileron; 85% rudder	CONSERVATIVE analytical allowance	S01/S02
VMC-05	Internal VMC gate	VMC <= 0.95 VR = 39.96 kn EAS	FINAL internal criterion	S02
VMC-06	VMC-A windmilling, CDf=0.45	38.97 kn EAS	CONDITIONAL PASS / thin margin	S02
VMC-07	VMC-A severe drag, CDf=0.60	39.37 kn EAS	CONDITIONAL PASS / thin margin	S02
VMC-08	VMC windmilling with 90% control travel	40.97 kn EAS	ROBUSTNESS FAIL by ~1.01 kn	S02
VMC-09	Hot-high VMC-A windmilling sensitivity	38.28 kn EAS / 40.78 kn TAS	SENSITIVITY; SL EAS governs	S02
VMC-10	Additional adverse powered-lift roll allowance at gate	~0.00206 windmill; ~0.00122 severe	OPEN / release-critical	S02

7. Landing Gear and Structural Headline Values

ID	Quantity	Controlled value	Use / status	Source
GEAR-01	Forward take-off static nose fraction / reaction	25.88%; RN = 4.084 kN	FINAL ground/control case	S02
GEAR-02	Forward landing static nose fraction / reaction	26.51%; RN = 3.950 kN	FINAL ground/control case	S02
GEAR-03	Light-forward nose-load range	25.9-26.5% static nose fraction	FINAL report range; handling validation remains	S02
GEAR-04	Landing-impact load factors	4.51 g limit; 6.76 g ultimate	FINAL analytical basis	S02

ID	Quantity	Controlled value	Use / status	Source
GEAR-05	Actual each-main landing demand	33.61 kN limit / 50.41 kN ultimate	PASS	S02
GEAR-06	Existing each-main design basis	44.2 kN limit / 66.3 kN ultimate	PASS; capacity/demand ~1.315	S02
GEAR-07	Worst forward nose-gear requirement	18.42 kN limit / 27.63 kN ultimate	OPEN supplier hardware rating	S02
GEAR-08	Tip-back angle, actual nominal aft CG	22.04 deg at FS 3.70312 m	PASS	S02
GEAR-09	Tip-back angle, U20 aft screen	17.32 deg at FS 3.83778 m	PASS screening boundary	S02
GEAR-10	Lateral-overturn angle, actual nominal / U20 aft	44.50 deg / 45.78 deg	PASS geometry screen	S02
STR-01	Primary wing-box material concept	Welded/reconsolidatable CF/PAEK box with metallic load-introduction fittings	FINAL structural concept; process qualification OPEN	S04/S07
STR-02	Primary wing-box mass comparison	190.1 kg CF/PAEK vs 225.0 kg matched aluminium	34.9 kg / 15.5% conceptual saving	S04/S07
STR-03	Installed wing-system comparison	~282.0 kg vs ~316.9 kg matched control	~11.0% conceptual saving	S04/S07

8. Performance Headline Values for the Final Report

These values remain the controlled report-level performance basis unless a direct report-integration arithmetic error is discovered. They are not flight-test or certification results; several depend on bounded high-lift, propeller and surface assumptions.

ID	Metric	Controlled result	Requirement / treatment	Source
PERF-01	Wing loading at MTOM	66.7 kg/m ²	<=90 kg/m ² - PASS	S04
PERF-02	Landing-config stall, unpowered	35.8 kn	<50 kn - PASS; depends on CL _{max} ,LDG=3.30 basis	S04
PERF-03	Take-off to 15 m	163 m SL / 172 m hot-high central / 179 m hot-high adverse	<200 m - PASS; powered-lift / thrust correlation remains open	S04
PERF-04	Landing from 15 m	148 m SL / 149 m hot-high at 1,900 kg MLM	<=150 m - PASS by ~1 m; reverse/braking/high-lift validation sensitive	S04
PERF-05	Cruise	130 kn at 2,600 m	Within 130-150 kn requirement	S04
PERF-06	Cruise shaft power	170.2 kW required vs 196 kW available	15.2% available-power margin	S04
PERF-07	Climb at 3,000 m	4.30 m/s	PASS	S04
PERF-08	Service ceiling	~8,750 m	>=3,000 m - PASS	S04

ID	Metric	Controlled result	Requirement / treatment	Source
PERF-09	505 km dispatch payload	563.2 kg people + payload with 109.8 kg fuel at MTOM	Delivered capability; not full RFP payload	S04
PERF-10	Six-occupant 505 km cargo	47.2 kg additional cargo	Delivered capability	S04
PERF-11	Full RFP payload mission	2,152.8 kg	FAIL MTOM by 152.8 kg	S04/S06
PERF-12	Extended ~886 km point	Mass-exchange sensitivity only	DO NOT present as delivered range; installed fuel volume not provided	S04

9. Validation, Uncertainty and Model-Boundary Register

Domain	Controlled evidence / uncertainty	Final report treatment
Aerodynamics - parasite drag	PDT CD0 = 0.018985 geometry, 0.021501 operational, 0.023067 adverse; report performance basis 0.0283	State 0.0283 as deliberately conservative. Never label it as directly validated by PDT.
Aerodynamics - wing lift slope	4.837 rad ⁻¹ analytical vs 4.88866 rad ⁻¹ MainWing VSPAERO (+1.07%)	PASS cross-check. Keep whole lifting-system 5.6188 separate.
Aerodynamics - induced drag	~5.25% k / ~5.54% e mesh sensitivity; Trefftz wake sensitivity ~1.2%	Carry uncertainty; use Trefftz CDi,w for convergence; do not tune to force convergence.
Aerodynamics - thick-body branch	Mixed thin/thick assembly remained physically invalid despite root repairs	REJECT production coefficients. Use thin lifting system + PDT + analytical fuselage correction.
High lift	CLmax,TO=2.76 and CLmax,LDG=3.30 are bounded analytical/design inputs	Do not claim FG8 validates multi-element CLmax or landing drag polar.
Stability / CG	Nominal and independent-95% aft screens pass; all-at-bounds aft case fails	Use nominal cloud; prohibit operation beyond FS 3.83778 pending CAD mass arms.
VMC - atmosphere	SL windmill 38.97 kn vs hot-high 38.28 kn EAS	Use explicit SL release atmosphere; delete ambiguous rho=1.168 headline result.
VMC - finite control	38.97 kn full travel; 40.97 kn at 90% travel	Conditional physical-stop pass; robustness fail remains visible.
VMC - propeller / powered roll	Failed-prop drag map and asymmetric powered-lift roll not frozen	Release-critical future validation; do not infer from rejected powered free-wake solver.
Forward-CG control	FG8.1 rotation, FG8.2 trim/flare, FG8.3 transition passed without powered-lift/thrust-line credit	Competition-level conceptual closure; dynamic 6-DoF go-around / flare remains future work.
Landing gear	Main-gear loads close; nose gear supplier rating and ground handling not frozen	Retain geometry; carry nose-gear hardware, steering, shimmy and tyre validation.
Structures	CF/PAEK mass benefit is conceptual	Retain claim with matched-baseline caveat; coupon/joint/hot-wet/fatigue/repair qualification remains open.
Energy storage	11 kWh / 8.8 kWh usable report basis	Carry cell/pack conformity, temperature, EOL capacity and thermal-runaway qualification.

Domain	Controlled evidence / uncertainty	Final report treatment
Simulation promises	No further VSPAERO/AVL/X-Plane activity required for the report sprint	Only list such work as future/higher-fidelity validation, not as an unexecuted submission promise.

10. Superseded / Unsafe Values - Mandatory Report-Replacement Register

Report item	Do not use as final statement	Controlled final statement
Geometry release	G1.1 / 9.70 m fuselage	G1.3 analysis geometry; fuselage length 10.750 m.
Wing lift-curve slope	5.6188 rad ⁻¹ described as the wing-only validation value	MainWing-only VSPAERO 4.88866 rad ⁻¹ vs analytical 4.837 rad ⁻¹ (+1.07%); 5.6188 is the complete lifting-system slope.
Parasite drag	CD0=0.0283 described as directly validated	PDT: 0.018985 geometry / 0.021501 operational / 0.023067 upper. Keep 0.0283 only as conservative performance basis.
Neutral point / CG	NP around FS 4.20-4.25 and acceptable CG FS 3.60-3.90	Corrected NP FS 4.02894; nominal loading cloud FS 3.34618-3.70312; U20 10%-MAC aft screen FS 3.83778.
VMC headline	38.73 / 39.09 kn using ambiguous rho=1.168 thrust state	Atmosphere-consistent sea-level-standard: 38.97 kn windmill, 39.37 kn severe; 40.97 kn at 90% travel.
Forward rotation	Forward-CG elevator / rotation evidence open	FG8.1 PASS: ~15 deg physical TE-up required; converged 20 deg case gives positive unload; >=17 deg TE-up design travel for 10% reserve.
Landing trim / flare	Forward-CG flare evidence open	FG8.2 PASS: ~4.08 deg TE-up steady trim; >5.9 deg additional tested TE-up movement.
Balked landing	No executed flap-transition evidence	FG8.3 PASS: quasi-steady 35 -> 20 -> 12 deg transition smooth with no pitch-control reversal.
Gear nose-load envelope	11-19.5% static nose-load band	Actual light-forward cases 25.9-26.5%, RN=3.95-4.08 kN; retain gear geometry with handling validation.
Gear vertical loads	Main-gear basis quoted without actual forward-case comparison	Actual 33.61/50.41 kN per main vs 44.2/66.3 kN existing basis - PASS.
Validation promises	AVL/X-Plane/external analyses presented as still required before report release	Replace with executed evidence register and explicit future/certification validation boundaries.

11. Source and Provenance Register

Source ID	Controlled evidence source	Principal use in this register
S01	Analysis Log.pdf - consolidated 01_CompGeom through final 05_Stability analysis record, 27 Aug 2026	Clean-aero convergence and polar, G5 derivatives, control derivatives, VMC derivation history, audit trail.
S02	Tyche_R2T1_Final_FG8_Closure_VMC_Atmosphere_Uncertainty_Register.docx - 27 Aug 2026	Latest forward-CG control and gear closure, VMC atmosphere repair, uncertainty register and report-replacement authority.

Source ID	Controlled evidence source	Principal use in this register
S03	Tyche_R2T1_SM6_3B_SM6_4_Integrated_CG_Closure_Pre_VM C.docx - 21 Aug 2026	Munk-Multhopp neutral point, mass-arm ledger, CG loading cloud and uncertainty screens.
S04	Tyche_R2-T1_Design_Report.pdf - current pre-reconciliation report baseline	RFP/mission basis, propulsion hardware, energy, high-lift design inputs, structural concept and performance headline results.
S05	Tyche_R2T1_G1_3_Analysis.vspscript / G1.3 analysis-ready geometry record	Final geometry, stations, wing and propeller layout, tail-arm correction.
S06	Tyche_R2_Study.pdf / Tyche_R2_Workbook.xlsx - R2 convergence basis	RFP common basis, architecture decision, mass-compliance closure and sensitivity context.
S07	Tyche_R2-T1_Design_Report_Chapters_4-6 / Chapters_7-9 source documents	Detailed propulsion, structural, high-lift, mass and sustainability report data.
S08	Tyche_R2T1_04_Clean_Aero_G1_4_to_G1_6_Analysis_Log.doc x	Production-model selection and rejection of mixed thin/thick body coefficients.

Release statement

CVR-1.0 is the numerical authority for the final Tyche R2-T1 competition report. Report drafting may compress, round or rearrange these values for readability, but it must not change their engineering meaning, precedence, qualification or pass/fail status. Any direct arithmetic or configuration-control discrepancy found during final integration must be corrected in this register first and then propagated consistently through the report.